

FederCare: AI Health Network

Database Design Document

Department of Computer Applications

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1. Table Name: login_credentials

Description: Stores login credentials and authentication details for all system users.

Attribute	Data Type	Constraints	Description
login_id	UUID	Primary Key, Not Null	Unique login identifier
email	Varchar(100)	Unique, Not Null	User email address
password_hash	Varchar(255)	Not Null	Encrypted password hash
role	Varchar(40)	Not Null	User role (super_admin/hospital_admin/doctor/patient/pharmacist/lab_tech/driver/vendor)
is_active	Boolean	Not Null	Account active status
is_approved	Boolean	Not Null	Account approval status
last_login	DateTime	Null	Last login timestamp
login_attempts	Integer	Not Null	Failed login attempts count
created_at	DateTime	Not Null	Record creation timestamp
updated_at	DateTime	Not Null	Record last update timestamp

2. Table Name: super_admin

Description: Stores super administrator profile details.

Attribute	Data Type	Constraints	Description
admin_id	UUID	Primary Key, Not Null	Unique admin identifier
login_id	UUID	Unique, FK to login_credentials	Reference to login credentials
full_name	Varchar(100)	Not Null	Administrator full name
phone	Varchar(15)	Not Null	Contact phone number
profile_photo	Varchar(500)	Not Null	Profile photo URL
created_at	DateTime	Not Null	Record creation timestamp
updated_at	DateTime	Not Null	Record last update timestamp

3. Table Name: role_permissions

Description: Stores module-level permissions for each user role.

Attribute	Data Type	Constraints	Description
permission_id	UUID	Primary Key, Not Null	Unique permission identifier
role	Varchar(40)	Not Null	User role name
module	Varchar(50)	Not Null	System module name

Attribute	Data Type	Constraints	Description
can_read	Boolean	Not Null	Read permission flag
can_write	Boolean	Not Null	Write permission flag
can_delete	Boolean	Not Null	Delete permission flag
created_at	DateTime	Not Null	Record creation timestamp

4. Table Name: login_sessions

Description: Tracks active login sessions and JWT tokens for security.

Attribute	Data Type	Constraints	Description
session_id	UUID	Primary Key, Not Null	Unique session identifier
login_id	UUID	FK to login_credentials	Reference to user login
jwt_token_hash	Varchar(255)	Not Null	Hashed JWT token
device_info	Text	Not Null	Client device information
ip_address	GenericIP	Null	Client IP address
expires_at	DateTime	Not Null	Session expiry timestamp
created_at	DateTime	Not Null	Session creation timestamp

5. Table Name: audit_logs

Description: Records all system actions for security and compliance auditing.

Attribute	Data Type	Constraints	Description
log_id	UUID	Primary Key, Not Null	Unique log identifier
login_id	UUID	FK to login_credentials, Null	User who performed action
action	Varchar(255)	Not Null	Action performed
module	Varchar(50)	Not Null	Module where action occurred
entity_type	Varchar(50)	Not Null	Type of entity affected
entity_id	UUID	Null	ID of affected entity
old_value	JSON	Null	Previous value before change
new_value	JSON	Null	New value after change
ip_address	GenericIP	Null	IP address of request
logged_at	DateTime	Not Null	Action timestamp

6. Table Name: notifications

Description: Stores system notifications sent to users across all roles.

Attribute	Data Type	Constraints	Description
notif_id	UUID	Primary Key, Not Null	Unique notification identifier
login_id	UUID	FK to login_credentials	Recipient user reference
title	Varchar(200)	Not Null	Notification title
message	Text	Not Null	Notification message content
notif_type	Varchar(30)	Not Null	Notification type (order/emergency/alert)
is_read	Boolean	Not Null	Read status flag
related_id	UUID	Null	Related entity identifier
created_at	DateTime	Not Null	Notification creation timestamp

7. Table Name: hospital_registrations

Description: Stores hospital registration and profile information.

Attribute	Data Type	Constraints	Description
hospital_id	UUID	Primary Key, Not Null	Unique hospital identifier
login_id	UUID	Unique, FK to login_credentials	Reference to login credentials
hospital_name	Varchar(200)	Not Null	Official hospital name
registration_no	Varchar(100)	Unique, Not Null	Government registration number
address	Text	Not Null	Hospital address
city	Varchar(100)	Not Null	City name
state	Varchar(100)	Not Null	State name
latitude	Decimal(9,7)	Null	GPS latitude coordinate
longitude	Decimal(9,7)	Null	GPS longitude coordinate
contact_phone	Varchar(15)	Not Null	Hospital contact number
contact_email	Varchar(100)	Not Null	Hospital contact email
doc_url	Varchar(500)	Not Null	Registration document URL
telemedicine_enabled	Boolean	Not Null	Telemedicine support flag
approval_status	Varchar(20)	Not Null	Approval status (pending/approved/rejected)
profile_photo	Varchar(500)	Not Null	Hospital photo URL
created_at	DateTime	Not Null	Record creation timestamp

Attribute	Data Type	Constraints	Description
updated_at	DateTime	Not Null	Record last update timestamp

8. Table Name: departments

Description: Stores hospital department information.

Attribute	Data Type	Constraints	Description
dept_id	UUID	Primary Key, Not Null	Unique department identifier
hospital_id	UUID	FK to hospital_registrations	Parent hospital reference
dept_name	Varchar(100)	Not Null	Department name
description	Text	Not Null	Department description
created_at	DateTime	Not Null	Record creation timestamp

9. Table Name: beds

Description: Stores hospital bed information and availability status.

Attribute	Data Type	Constraints	Description
bed_id	UUID	Primary Key, Not Null	Unique bed identifier
hospital_id	UUID	FK to hospital_registrations	Parent hospital reference
bed_type	Varchar(50)	Not Null	Bed type (general/ICU/emergency)
ward_name	Varchar(100)	Not Null	Ward name
status	Varchar(20)	Not Null	Bed status (available/occupied/reserved)
reserved_for	UUID	FK to patient_registrations, Null	Reserved patient reference
reserved_at	DateTime	Null	Reservation timestamp
admitted_at	DateTime	Null	Patient admission timestamp
updated_at	DateTime	Not Null	Last update timestamp

10. Table Name: hospital_inventory

Description: Tracks medical equipment and supply inventory for hospitals.

Attribute	Data Type	Constraints	Description
inventory_id	UUID	Primary Key, Not Null	Unique inventory identifier
hospital_id	UUID	FK to hospital_registrations	Parent hospital reference

Attribute	Data Type	Constraints	Description
item_name	Varchar(200)	Not Null	Item name
category	Varchar(50)	Not Null	Item category
quantity	Integer	Not Null	Available quantity
unit	Varchar(20)	Not Null	Unit of measurement
reorder_level	Integer	Not Null	Minimum reorder threshold
last_restocked	DateTime	Null	Last restock timestamp
maintenance_due	Date	Null	Next maintenance date

11. Table Name: hospital_patients

Description: Stores patient records added by hospitals for federated learning training data.

Attribute	Data Type	Constraints	Description
patient_id	UUID	Primary Key, Not Null	Unique record identifier
hospital_id	UUID	FK to hospital_registrations	Parent hospital reference
added_by	UUID	FK to login_credentials, Null	Staff who added record
full_name	Varchar(100)	Not Null	Patient full name
age	Integer	Not Null	Patient age
gender	Varchar(10)	Not Null	Patient gender
blood_group	Varchar(5)	Not Null	Patient blood group
symptoms	JSON	Not Null	Symptoms list
diagnosis	Varchar(200)	Not Null	Medical diagnosis
visit_date	Date	Not Null	Hospital visit date
notes	Text	Not Null	Clinical notes
created_at	DateTime	Not Null	Record creation timestamp

12. Table Name: patient_registrations

Description: Stores registered patient profile and health information.

Attribute	Data Type	Constraints	Description
patient_id	UUID	Primary Key, Not Null	Unique patient identifier
login_id	UUID	Unique, FK to login_credentials	Reference to login credentials

Attribute	Data Type	Constraints	Description
full_name	Varchar(100)	Not Null	Patient full name
dob	Date	Not Null	Date of birth
gender	Varchar(10)	Not Null	Patient gender
blood_group	Varchar(5)	Not Null	Blood group
height_cm	Decimal	Null	Height in centimeters
weight_kg	Decimal	Null	Weight in kilograms
bmi	Decimal	Null	Body Mass Index
address	Text	Not Null	Residential address
emergency_contact	Varchar(15)	Not Null	Emergency contact number
qr_code_url	Varchar(500)	Not Null	EHR QR code URL
lifestyle_data	JSON	Not Null	Lifestyle and health habits data
created_at	DateTime	Not Null	Record creation timestamp
updated_at	DateTime	Not Null	Record last update timestamp

13. Table Name: ehr_records

Description: Stores Electronic Health Records for patients.

Attribute	Data Type	Constraints	Description
record_id	UUID	Primary Key, Not Null	Unique record identifier
patient_id	UUID	FK to patient_registrations	Patient reference
added_by	UUID	FK to login_credentials, Null	Staff who added record
record_type	Varchar(50)	Not Null	Record type (prescription/lab/diagnosis)
title	Varchar(200)	Not Null	Record title
content	Text	Not Null	Record content
file_url	Varchar(500)	Not Null	Attached file URL
is_sensitive	Boolean	Not Null	Sensitive data flag
recorded_at	DateTime	Not Null	Record timestamp

14. Table Name: allergies

Description: Stores patient allergy information for medical safety.

Attribute	Data Type	Constraints	Description
allergy_id	UUID	Primary Key, Not Null	Unique allergy identifier
patient_id	UUID	FK to patient_registrations	Patient reference
allergen	Varchar(100)	Not Null	Allergen name
reaction	Text	Not Null	Reaction description
severity	Varchar(20)	Not Null	Severity level (mild/moderate/severe)
noted_by	UUID	FK to doctor_registrations, Null	Doctor who noted allergy
noted_at	DateTime	Not Null	Date allergy was noted

15. Table Name: ehr_consent_log

Description: Tracks patient consent for EHR data access by doctors.

Attribute	Data Type	Constraints	Description
consent_id	UUID	Primary Key, Not Null	Unique consent identifier
patient_id	UUID	FK to patient_registrations	Patient reference
accessed_by	UUID	FK to login_credentials	Staff who accessed EHR
access_type	Varchar(50)	Not Null	Access type (view/download)
data_shared	JSON	Not Null	List of shared data fields
consent_given	Boolean	Not Null	Consent status flag
expires_at	DateTime	Null	Consent expiry timestamp
accessed_at	DateTime	Not Null	Access timestamp

16. Table Name: risk_assessments

Description: Stores AI-generated health risk assessment results for patients.

Attribute	Data Type	Constraints	Description
risk_id	UUID	Primary Key, Not Null	Unique assessment identifier
patient_id	UUID	FK to patient_registrations	Patient reference
diabetes_risk	Decimal	Null	Diabetes risk percentage
heart_risk	Decimal	Null	Heart disease risk percentage
hypertension_risk	Decimal	Null	Hypertension risk percentage
risk_level	Varchar(20)	Not Null	Overall risk level (low/medium/high)
recommendations	Text	Not Null	AI recommendations

Attribute	Data Type	Constraints	Description
alert_sent	Boolean	Not Null	Alert notification sent flag
assessed_at	DateTime	Not Null	Assessment timestamp

17. Table Name: patient_complaints

Description: Stores patient complaints about doctors, hospitals or vendors.

Attribute	Data Type	Constraints	Description
complaint_id	UUID	Primary Key, Not Null	Unique complaint identifier
patient_id	UUID	FK to patient_registrations	Complainant patient reference
complaint_type	Varchar(30)	Not Null	Complaint type (doctor/vendor/hospital)
doctor_id	UUID	FK to doctor_registrations, Null	Complained doctor reference
hospital_id	UUID	FK to hospital_registrations, Null	Complained hospital reference
vendor_id	UUID	FK to vendor_registrations, Null	Complained vendor reference
subject	Varchar(200)	Not Null	Complaint subject
description	Text	Not Null	Detailed complaint description
status	Varchar(20)	Not Null	Status (pending/reviewed/resolved)
admin_response	Text	Not Null	Admin response to complaint
created_at	DateTime	Not Null	Complaint submission timestamp
updated_at	DateTime	Not Null	Last update timestamp

18. Table Name: lab_test_orders

Description: Stores patient lab test bookings and results.

Attribute	Data Type	Constraints	Description
order_id	UUID	Primary Key, Not Null	Unique order identifier
patient_id	UUID	FK to patient_registrations	Patient reference
hospital_id	UUID	FK to hospital_registrations, Null	Hospital reference
doctor_id	UUID	FK to doctor_registrations, Null	Referring doctor reference
tests	JSON	Not Null	List of ordered tests
total_fee	Decimal	Not Null	Total test fee

Attribute	Data Type	Constraints	Description
appointment_date	Date	Null	Scheduled appointment date
appointment_time	Time	Null	Scheduled appointment time
status	Varchar(20)	Not Null	Order status
payment_status	Varchar(20)	Not Null	Payment status
razorpay_order_id	Varchar(100)	Not Null	Razorpay order ID
razorpay_payment_id	Varchar(100)	Not Null	Razorpay payment ID
report_url	Varchar(500)	Not Null	Lab report URL
report_results	JSON	Not Null	Test results data
abnormal_flags	JSON	Not Null	Abnormal result flags
notes	Text	Not Null	Additional notes
ordered_at	DateTime	Not Null	Order creation timestamp

19. Table Name: ehr_images

Description: Stores medical images uploaded to patient EHR wallet.

Attribute	Data Type	Constraints	Description
image_id	UUID	Primary Key, Not Null	Unique image identifier
patient_id	UUID	FK to patient_registrations	Patient reference
image_type	Varchar(30)	Not Null	Image type (X-Ray/MRI/CT/Ultrasound)
image_url	Varchar(500)	Not Null	Image storage URL
title	Varchar(200)	Not Null	Image title
description	Text	Not Null	Image description
hospital_name	Varchar(200)	Not Null	Hospital where scan was done
scan_date	Date	Null	Date of scan
uploaded_by	UUID	FK to login_credentials, Null	Uploader reference
uploaded_at	DateTime	Not Null	Upload timestamp

20. Table Name: doctor_registrations

Description: Stores registered doctor profile and professional details.

Attribute	Data Type	Constraints	Description
doctor_id	UUID	Primary Key, Not Null	Unique doctor identifier

Attribute	Data Type	Constraints	Description
login_id	UUID	Unique, FK to login_credentials	Reference to login credentials
hospital_id	UUID	FK to hospital_registrations	Associated hospital reference
dept_id	UUID	FK to departments, Null	Department reference
full_name	Varchar(100)	Not Null	Doctor full name
specialization	Varchar(100)	Not Null	Medical specialization
license_no	Varchar(50)	Unique, Not Null	Medical license number
experience_years	Integer	Not Null	Years of experience
consultation_fee	Decimal	Not Null	Consultation fee amount
profile_photo	Varchar(500)	Not Null	Profile photo URL
is_online	Boolean	Not Null	Online availability status
approval_status	Varchar(20)	Not Null	Account approval status
created_at	DateTime	Not Null	Record creation timestamp
updated_at	DateTime	Not Null	Record last update timestamp

21. Table Name: doctor_slots

Description: Stores doctor consultation time slots for appointment booking.

Attribute	Data Type	Constraints	Description
slot_id	UUID	Primary Key, Not Null	Unique slot identifier
doctor_id	UUID	FK to doctor_registrations	Doctor reference
slot_date	Date	Not Null	Appointment date
start_time	Time	Not Null	Slot start time
end_time	Time	Not Null	Slot end time
consult_type	Varchar(20)	Not Null	Consultation type (online/offline)
is_booked	Boolean	Not Null	Booking status flag
created_at	DateTime	Not Null	Record creation timestamp

22. Table Name: consultations

Description: Stores patient-doctor consultation records including video call details.

Attribute	Data Type	Constraints	Description
consultation_id	UUID	Primary Key, Not Null	Unique consultation identifier

Attribute	Data Type	Constraints	Description
patient_id	UUID	FK to patient_registrations	Patient reference
doctor_id	UUID	FK to doctor_registrations	Doctor reference
slot_id	UUID	FK to doctor_slots, Null	Booked slot reference
jitsi_room_id	Varchar(200)	Not Null	Jitsi Meet room identifier
status	Varchar(20)	Not Null	Status (scheduled/active/completed/cancelled)
ai_suggestions	JSON	Not Null	AI diagnosis suggestions
doctor_notes	Text	Not Null	Doctor consultation notes
final_diagnosis	Text	Not Null	Final diagnosis text
to_emergency	Boolean	Not Null	Emergency escalation flag
razorpay_order_id	Varchar(100)	Not Null	Razorpay order ID
payment_status	Varchar(20)	Not Null	Payment status
started_at	DateTime	Null	Consultation start timestamp
ended_at	DateTime	Null	Consultation end timestamp
created_at	DateTime	Not Null	Record creation timestamp

23. Table Name: prescriptions

Description: Stores doctor-issued prescriptions with medicine details.

Attribute	Data Type	Constraints	Description
prescription_id	UUID	Primary Key, Not Null	Unique prescription identifier
doctor_id	UUID	FK to doctor_registrations	Prescribing doctor reference
patient_id	UUID	FK to patient_registrations	Patient reference
consultation_id	UUID	FK to consultations, Null	Related consultation reference
medicines	JSON	Not Null	List of prescribed medicines with dosage
diagnosis	Text	Not Null	Diagnosis for prescription
instructions	Text	Not Null	Usage instructions
is_verified	Boolean	Not Null	Prescription verification status
valid_until	Date	Null	Prescription validity date
pdf_url	Varchar(500)	Not Null	Generated PDF URL
created_at	DateTime	Not Null	Record creation timestamp

24. Table Name: pharmacist_registrations

Description: Stores registered pharmacist and pharmacy details.

Attribute	Data Type	Constraints	Description
pharmacist_id	UUID	Primary Key, Not Null	Unique pharmacist identifier
login_id	UUID	Unique, FK to login_credentials	Reference to login credentials
pharmacy_name	Varchar(200)	Not Null	Pharmacy name
license_no	Varchar(50)	Unique, Not Null	Pharmacy license number
full_name	Varchar(100)	Not Null	Pharmacist full name
address	Text	Not Null	Pharmacy address
latitude	Decimal	Null	GPS latitude coordinate
longitude	Decimal	Null	GPS longitude coordinate
approval_status	Varchar(20)	Not Null	Account approval status
created_at	DateTime	Not Null	Record creation timestamp
updated_at	DateTime	Not Null	Record last update timestamp

25. Table Name: medicine_orders

Description: Stores patient medicine orders with prescription verification and delivery tracking.

Attribute	Data Type	Constraints	Description
med_order_id	UUID	Primary Key, Not Null	Unique order identifier
patient_id	UUID	FK to patient_registrations	Patient reference
pharmacist_id	UUID	FK to pharmacist_registrations, Null	Pharmacist reference
prescription_id	UUID	FK to prescriptions, Null	Related prescription reference
medicines	JSON	Not Null	Ordered medicines list
total_amount	Decimal	Not Null	Total order amount
payment_status	Varchar(20)	Not Null	Payment status
razorpay_order_id	Varchar(100)	Not Null	Razorpay order ID
delivery_address	Text	Not Null	Delivery address
order_status	Varchar(30)	Not Null	Order status
prescription_url	Varchar(500)	Not Null	Prescription file URL
prescription_verified	Boolean	Not Null	Prescription verification flag

Attribute	Data Type	Constraints	Description
requires_prescription	Boolean	Not Null	Prescription requirement flag
payment_enabled	Boolean	Not Null	Payment enabled flag
delivery_otp	Varchar(6)	Not Null	Delivery OTP code
otp_expiry	DateTime	Null	OTP expiry timestamp
otp_verified	Boolean	Not Null	OTP verification flag
estimated_delivery_days	Integer	Not Null	Estimated delivery days
dispatched_at	DateTime	Null	Dispatch timestamp
delivered_at	DateTime	Null	Delivery timestamp
status_history	JSON	Not Null	Order status history log
ordered_at	DateTime	Not Null	Order creation timestamp
updated_at	DateTime	Not Null	Last update timestamp

26. Table Name: **pharmacy_inventory**

Description: Stores pharmacy medicine inventory with stock and expiry details.

Attribute	Data Type	Constraints	Description
inventory_id	UUID	Primary Key, Not Null	Unique inventory identifier
pharmacy_id	UUID	FK to pharmacist_registrations	Pharmacy reference
medicine_name	Varchar(200)	Not Null	Medicine name
generic_name	Varchar(200)	Not Null	Generic medicine name
category	Varchar(20)	Not Null	Category (tablet/syrup/injection/cream)
description	Text	Not Null	Medicine description
price_per_unit	Decimal	Not Null	Price per unit
unit	Varchar(20)	Not Null	Unit type
stock_quantity	Integer	Not Null	Available stock count
reorder_level	Integer	Not Null	Minimum reorder threshold
requires_prescription	Boolean	Not Null	Prescription requirement flag
medicine_image	FileField	Null	Medicine image file
manufacturer	Varchar(200)	Not Null	Manufacturer name
expiry_date	Date	Null	Medicine expiry date

Attribute	Data Type	Constraints	Description
is_available	Boolean	Not Null	Availability flag
created_at	DateTime	Not Null	Record creation timestamp
updated_at	DateTime	Not Null	Record last update timestamp

27. Table Name: lab_tech_registrations

Description: Stores registered laboratory technician profile details.

Attribute	Data Type	Constraints	Description
lab_tech_id	UUID	Primary Key, Not Null	Unique lab tech identifier
login_id	UUID	Unique, FK to login_credentials	Reference to login credentials
hospital_id	UUID	FK to hospital_registrations	Associated hospital reference
full_name	Varchar(100)	Not Null	Lab technician full name
qualification	Varchar(100)	Not Null	Educational qualification
specialization	Varchar(100)	Not Null	Area of specialization
phone	Varchar(15)	Not Null	Contact phone number
approval_status	Varchar(20)	Not Null	Account approval status
created_at	DateTime	Not Null	Record creation timestamp
updated_at	DateTime	Not Null	Record last update timestamp

28. Table Name: lab_orders

Description: Stores lab test orders created by doctors during consultations.

Attribute	Data Type	Constraints	Description
order_id	UUID	Primary Key, Not Null	Unique order identifier
doctor_id	UUID	FK to doctor_registrations	Ordering doctor reference
patient_id	UUID	FK to patient_registrations	Patient reference
lab_tech_id	UUID	FK to lab_tech_registrations, Null	Assigned lab tech reference
tests_ordered	JSON	Not Null	List of tests ordered
priority	Varchar(20)	Not Null	Priority level (STAT/URGENT/NORMAL)
status	Varchar(20)	Not Null	Order status
notes	Text	Not Null	Additional notes

Attribute	Data Type	Constraints	Description
payment_status	Varchar(20)	Not Null	Payment status
ordered_at	DateTime	Not Null	Order creation timestamp
updated_at	DateTime	Not Null	Last update timestamp

29. Table Name: lab_reports

Description: Stores completed lab test reports with results and AI analysis.

Attribute	Data Type	Constraints	Description
report_id	UUID	Primary Key, Not Null	Unique report identifier
order_id	UUID	FK to lab_orders	Related lab order reference
patient_id	UUID	FK to patient_registrations	Patient reference
results	JSON	Not Null	Test results data
report_file_url	Varchar(500)	Not Null	Report file URL
abnormal_flags	JSON	Not Null	Abnormal result indicators
ai_analysis	Text	Not Null	AI-generated analysis
saved_to_ehr	Boolean	Not Null	EHR save status flag
uploaded_at	DateTime	Not Null	Report upload timestamp

30. Table Name: ambulance_driver_registrations

Description: Stores registered ambulance driver profile details.

Attribute	Data Type	Constraints	Description
driver_id	UUID	Primary Key, Not Null	Unique driver identifier
login_id	UUID	Unique, FK to login_credentials	Reference to login credentials
hospital_id	UUID	FK to hospital_registrations	Associated hospital reference
full_name	Varchar(100)	Not Null	Driver full name
license_no	Varchar(50)	Unique, Not Null	Driving license number
phone	Varchar(15)	Not Null	Contact phone number
is_available	Boolean	Not Null	Availability status flag
approval_status	Varchar(20)	Not Null	Account approval status
created_at	DateTime	Not Null	Record creation timestamp
updated_at	DateTime	Not Null	Record last update timestamp

31. Table Name: ambulances

Description: Stores ambulance vehicle details and real-time GPS location.

Attribute	Data Type	Constraints	Description
ambulance_id	UUID	Primary Key, Not Null	Unique ambulance identifier
hospital_id	UUID	FK to hospital_registrations	Associated hospital reference
driver_id	UUID	FK to ambulance_driver_registrations, Null	Assigned driver reference
vehicle_no	Varchar(20)	Unique, Not Null	Vehicle registration number
ambulance_type	Varchar(20)	Not Null	Type (basic/advanced/ICU)
equipment_list	JSON	Not Null	Onboard equipment list
is_available	Boolean	Not Null	Availability status flag
current_lat	Decimal(10,7)	Null	Current GPS latitude
current_lng	Decimal(10,7)	Null	Current GPS longitude
updated_at	DateTime	Not Null	Last update timestamp

32. Table Name: emergency_requests

Description: Stores patient emergency SOS requests with location and severity.

Attribute	Data Type	Constraints	Description
emergency_id	UUID	Primary Key, Not Null	Unique emergency identifier
patient_id	UUID	FK to patient_registrations	Patient reference
triage_id	UUID	FK to triage_sessions, Null	Related triage session
patient_lat	Decimal(10,7)	Not Null	Patient GPS latitude
patient_lng	Decimal(10,7)	Not Null	Patient GPS longitude
severity	Varchar(20)	Not Null	Severity level (low/moderate/high/critical)
status	Varchar(20)	Not Null	Status (pending/dispatched/completed)
assigned_hospital_id	UUID	FK to hospital_registrations, Null	Nearest hospital reference
assigned_bed_id	UUID	FK to beds, Null	Assigned bed reference
created_at	DateTime	Not Null	Emergency creation timestamp
updated_at	DateTime	Not Null	Last update timestamp

33. Table Name: ambulance_dispatch

Description: Stores ambulance dispatch records linking emergencies to ambulances.

Attribute	Data Type	Constraints	Description
dispatch_id	UUID	Primary Key, Not Null	Unique dispatch identifier
emergency_id	UUID	FK to emergency_requests	Emergency reference
ambulance_id	UUID	FK to ambulances	Assigned ambulance reference
dispatch_status	Varchar(30)	Not Null	Status (dispatched/en_route/arrived/completed)
eta_minutes	Integer	Null	Estimated arrival time in minutes
route_data	JSON	Not Null	Route waypoints data
dispatched_at	DateTime	Not Null	Dispatch timestamp
arrived_at	DateTime	Null	Arrival timestamp
completed_at	DateTime	Null	Trip completion timestamp

34. Table Name: vendor_registrations

Description: Stores registered medical equipment vendor details.

Attribute	Data Type	Constraints	Description
vendor_id	UUID	Primary Key, Not Null	Unique vendor identifier
login_id	UUID	Unique, FK to login_credentials	Reference to login credentials
company_name	Varchar(200)	Not Null	Company name
tax_id	Varchar(50)	Unique, Not Null	Tax identification number
contact_name	Varchar(100)	Not Null	Primary contact person name
phone	Varchar(15)	Not Null	Contact phone number
business_license_url	Varchar(500)	Not Null	Business license document URL
certifications	JSON	Not Null	List of certifications
approval_status	Varchar(20)	Not Null	Account approval status
created_at	DateTime	Not Null	Record creation timestamp
updated_at	DateTime	Not Null	Record last update timestamp

35. Table Name: equipment_catalog

Description: Stores medical equipment listed by vendors for hospital purchase.

Attribute	Data Type	Constraints	Description
product_id	UUID	Primary Key, Not Null	Unique product identifier

Attribute	Data Type	Constraints	Description
vendor_id	UUID	FK to vendor_registrations	Vendor reference
product_name	Varchar(200)	Not Null	Equipment product name
category	Varchar(50)	Not Null	Equipment category
specifications	JSON	Not Null	Technical specifications
price	Decimal	Not Null	Product price
stock_qty	Integer	Not Null	Available stock quantity
image_url	Varchar(500)	Not Null	Product image URL
listed_at	DateTime	Not Null	Listing creation timestamp
updated_at	DateTime	Not Null	Last update timestamp

36. Table Name: equipment_orders

Description: Stores hospital equipment orders with payment and OTP delivery tracking.

Attribute	Data Type	Constraints	Description
eq_order_id	UUID	Primary Key, Not Null	Unique order identifier
hospital_id	UUID	FK to hospital_registrations	Ordering hospital reference
vendor_id	UUID	FK to vendor_registrations	Vendor reference
product_id	UUID	FK to equipment_catalog	Ordered product reference
quantity	Integer	Not Null	Ordered quantity
total_price	Decimal	Not Null	Total order price
order_status	Varchar(30)	Not Null	Order status
razorpay_order_id	Varchar(100)	Not Null	Razorpay order ID
payment_status	Varchar(20)	Not Null	Payment status
tracking_info	Text	Not Null	Delivery tracking information
delivery_otp	Varchar(6)	Not Null	Delivery OTP code
otp_expiry	DateTime	Null	OTP expiry timestamp
otp_verified	Boolean	Not Null	OTP verification flag
estimated_delivery_days	Integer	Not Null	Estimated delivery days
dispatched_at	DateTime	Null	Dispatch timestamp
delivered_at	DateTime	Null	Delivery timestamp

Attribute	Data Type	Constraints	Description
status_history	JSON	Not Null	Order status history log
ordered_at	DateTime	Not Null	Order creation timestamp
updated_at	DateTime	Not Null	Last update timestamp

37. Table Name: triage_sessions

Description: Stores AI-powered triage session results for patient symptom checking.

Attribute	Data Type	Constraints	Description
triage_id	UUID	Primary Key, Not Null	Unique triage identifier
patient_id	UUID	FK to patient_registrations	Patient reference
symptoms_input	JSON	Not Null	Patient reported symptoms
predicted_diseases	JSON	Not Null	AI predicted disease list with confidence
confidence_score	Decimal	Null	Overall prediction confidence score
severity	Varchar(20)	Not Null	Assessed severity level
model_version	Varchar(50)	Not Null	AI model version used
emergency_triggered	Boolean	Not Null	Emergency SOS trigger flag
recommendation	Text	Not Null	AI recommendation text
created_at	DateTime	Not Null	Session creation timestamp

38. Table Name: fl_global_models

Description: Stores federated learning global model versions and performance metrics.

Attribute	Data Type	Constraints	Description
model_id	UUID	Primary Key, Not Null	Unique model identifier
version	Varchar(50)	Unique, Not Null	Model version string
weights_file_url	Varchar(500)	Not Null	Model weights file URL
accuracy	Decimal	Null	Model accuracy percentage
hospitals_count	Integer	Not Null	Number of participating hospitals
aggregation_algo	Varchar(50)	Not Null	Aggregation algorithm (FedAvg)
is_active	Boolean	Not Null	Active model flag
privacy_epsilon	Decimal	Null	Differential privacy epsilon value
created_at	DateTime	Not Null	Model creation timestamp

39. Table Name: fl_rounds

Description: Stores federated learning training round details and status.

Attribute	Data Type	Constraints	Description
round_id	UUID	Primary Key, Not Null	Unique round identifier
model_id	UUID	FK to fl_global_models	Parent model reference
round_number	Integer	Not Null	Sequential round number
status	Varchar(20)	Not Null	Round status (pending/active/completed)
hospitals_invited	Integer	Not Null	Number of invited hospitals
hospitals_completed	Integer	Not Null	Number of completed submissions
global_loss	Decimal	Null	Global model loss value
started_at	DateTime	Null	Round start timestamp
completed_at	DateTime	Null	Round completion timestamp
round_deadline	DateTime	Null	Submission deadline
min_hospitals_thresh old	Integer	Not Null	Minimum required hospitals
auto_aggregated	Boolean	Not Null	Auto aggregation flag
reminder_sent	Boolean	Not Null	Reminder notification sent flag
created_at	DateTime	Not Null	Record creation timestamp

40. Table Name: fl_hospital_weights

Description: Stores individual hospital model weight submissions for federated learning.

Attribute	Data Type	Constraints	Description
weight_id	UUID	Primary Key, Not Null	Unique weight identifier
round_id	UUID	FK to fl_rounds	FL round reference
hospital_id	UUID	FK to hospital_registrations	Submitting hospital reference
weights_file_url	Varchar(500)	Not Null	Weight file URL
local_accuracy	Decimal	Null	Local model accuracy
local_loss	Decimal	Null	Local model loss
training_samples	Integer	Not Null	Number of training samples used
noise_added	Boolean	Not Null	Differential privacy noise flag
submitted_at	DateTime	Not Null	Submission timestamp

41. Table Name: epidemic_trends

Description: Stores epidemic detection and disease trend monitoring data.

Attribute	Data Type	Constraints	Description
trend_id	UUID	Primary Key, Not Null	Unique trend identifier
disease_name	Varchar(100)	Not Null	Disease name
region	Varchar(100)	Not Null	Affected region
case_count	Integer	Not Null	Number of reported cases
spike_detected	Boolean	Not Null	Epidemic spike detection flag
heatmap_data	JSON	Not Null	Geographic heatmap data
alert_level	Varchar(20)	Not Null	Alert level (low/medium/high/critical)
recorded_date	Date	Not Null	Data recording date